Sterile Neutrino Dark Matter: Joint BBN and X-ray Constraints 2024–2026

HEP-Genesis-Agent session record
Project: ramp-research-sterile-bbn-xray

2026-11-02

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Abstract

A multi-wave survey of 2024–2026 arXiv work on keV sterile-neutrino dark matter was combined with direct calculations from a sterile-neutrino BBN/X-ray bridge server. BBN observables Y_p, D/H and $\Delta N_{\rm eff}$ were computed over $m_s = 1$ eV to 100 keV and $\sin^2 2\theta = 10^{-15}$ to 10^{-1} for Dodelson–Widrow (DW) and Shi–Fuller (SF) mechanisms, together with radiative decay predictions $E_\gamma = m_s/2$, Γ and lifetime. The server baselines $Y_p = 0.2449 \pm 0.004$, D/H$= 2.547 \pm 0.025 \times 10^{-5}$, $N_{\rm eff,CMB} = 2.99 \pm 0.17$ give a BBN 95% limit $\Delta N_{\rm eff} < 0.548$ ($\Delta\chi^2 > 3.84$). BBN alone allows all keV DW points at X-ray relevant mixings, while built-in NuSTAR M31 and XMM blank-sky limits exclude DW $\Omega_s h^2 = 0.12$ at $m_s = 6$–20 keV. Non-minimal production with large lepton asymmetry $L \gtrsim 0.5$–1 at $T > 20$ MeV or inflaton-decay branching ratio $\lesssim 10^{-4}$ is required to reach the dark-matter abundance at $\sin^2 2\theta \lesssim 10^{-12}$–$10^{-14}$ where X-ray limits weaken. Structure-formation bounds $m_{\rm th} > 6.5$ keV (satellites) and > 6.1 keV (lensing) translate via $m_s = 4.43\ {\rm keV}(m_{\rm th}/{\rm keV})^{4/3}$ to $m_s > 53.7$ keV and > 49 keV for DW, but are colder for resonant and reheating production. The DW normalization in the server is a factor ~ 12 below the published Abazajian 2017 fit, shifting the $\Omega_s h^2 = 0.12$ contour. No 2024–2026 eROSITA or INTEGRAL/SPI primary line-search paper was retrieved; several X-ray and BBN robustness checks remained incomplete due to tool-budget exhaustion and DNS failures.

1 Introduction

The session addressed a single overarching question posed repeatedly with evolving emphasis: survey recent arXiv work from 2024 to 2026 on sterile-neutrino dark matter and combine BBN constraints computed with the sterile-nu tools with X-ray decay-line constraints to map viable $(m_s, \sin^2 2\theta)$ regions per production mechanism, including structure-formation bounds.

The initial request required using the sterile-neutrino server to compute Y_p, D/H and $\Delta N_{\rm eff}$ together with relic-density constraints across the $(m_s, \sin^2 2\theta)$ plane, quantifying X-ray line energies and fluxes and comparing them with 2024–2026 limits from NuSTAR, INTEGRAL/SPI, Chandra, eROSITA and XRISM, surveying production-mechanism literature on Shi–Fuller resonant production with lepton asymmetry, reheating production and scalar decays, and synthesizing a per-mechanism viability table with provenance and robustness checks. Subsequent continuations narrowed the focus to the two weakest claims identified by the research board: the quantitative X-ray limit table at $m_s = 7.1, 10, 20, 40, 50$ keV and the production-mechanism/structure-formation

mapping, including the factor $\sim$ 10–17 DW relic-density normalization dispute and the Viel et al. $m_{\rm th} \to m_s$ translation, and to BBN robustness under different Y_p/D/H baselines and thermalization fractions. Later waves explicitly requested deepening the BBN L-dependence and the X-ray exclusion of DW-motivated dark matter versus alternative production and field-of-view systematics, and verification of 2024–2026 $m_{\rm th}$ bounds. The scope remained keV sterile neutrinos as warm/cold dark matter, with eV-scale cross-checks for the BEST short-baseline anomaly, and explicitly excluded MeV+ heavy neutral leptons where decay injection is not modeled.

2 Methods

All quantitative BBN and X-ray numbers were produced with the Sterile Neutrino → BBN/X-ray Bridge Server via `serve-sbbn.sh` on local dispatch, complemented by `hep-research-aid` and `local` web and arXiv search tools. The server provides two skills, of which `sterile-nu-bbn-constraints` was loaded. BBN abundances are from a fit to PRyMordial calibrated to $< 0.3\%$, with the only BBN input being $\Delta N_{\rm eff}$. The fit is valid for $N_{\rm eff}$ in $[2.044, 6.044]$, $\omega_b h^2$ in $[0.019, 0.0257]$ and τ_n in $[866.4, 890.4]$ s, and for $m_s \lesssim 100$ keV where the sterile is relativistic at BBN. For $m_s \gtrsim 1$ MeV the server flags `bbn_treatment_valid=false` and decay-product injection is not modeled. Exclusion is $\Delta\chi^2 > 3.84$ versus the Standard Model at 95% for one degree of freedom. Observational baselines adopted by the server are $Y_p = 0.2449 \pm 0.004$ from Aver, Olive and Skillman JCAP 07 (2021) 029, D/H$= 2.547 \pm 0.025 \times 10^{-5}$ from Cooke, Pettini and Steidel ApJ 855 (2018) 102, $N_{\rm eff,CMB} = 2.99 \pm 0.17$ from Planck 2018 TT,TE,EE+lowE+lensing+BAO, and CMB-S4 forecast 3.044 ± 0.03, with Standard Model $N_{\rm eff} = 3.044$, $Y_p = 0.24682$, D/H$= 2.4433 \times 10^{-5}$ and $\chi^2 = 2.77$ at the null. Mechanisms are DW non-resonant and SF resonant, where SF $\Delta N_{\rm eff}$ is a heuristic 0.1× DW and lepton asymmetry L is capped at 0.1 with no ξ_e modeling. X-ray limits in the server are ten tabulated NuSTAR/XMM limits across five targets (Perseus, M31, Milky Way center, Coma, stacked galaxies) including NuSTAR M31 Roach et al. 2020 and XMM blank-sky Dessert et al. 2020, evaluated for whole-halo $f_{\rm DM} = 1$ with no field-of-view cut. Radiative decay follows $\Gamma = 1.38 \times 10^{-32}\,{\rm s}^{-1}\,(\sin^2 2\theta/10^{-10})(m_s/{\rm keV})^5$, equivalently $\Gamma = 1.38 \times 10^{-29}\,{\rm s}^{-1}\,(\sin^2 2\theta/10^{-7})(m_s/{\rm keV})^5$, with $E_\gamma = m_s/2$.

Computations executed included `predict_sterile_nu` at single points, `predict_sterile_nu_grid` over masses $[1, 3, 7.1, 10, 20, 50, 100]$ keV and mixings 10^{-15} to 10^{-6} and over 1 eV to 100 keV $\times 10^{-13}$ to 0.1, `predict_sterile_nu_xray_line` for E_γ, Γ, lifetime and fluxes, `scan_sterile_nu_parameters` for 400 to 10000 point scans, `check_sterile_nu_sbl_anomaly` for the BEST point, and `plot_sterile_nu_bbn_impact` and `plot_sterile_nu_exclusion` for BBN versus $N_{\rm eff}$ and exclusion-plane figures. Literature searches used `search_arxiv_abstracts`, `search_arxiv`, `get_arxiv_metadata`, `download_arxiv_full_text`, `read_arxiv_chunk`, `web_search`, `get_paper_figures` and `cite`. The supplement holds the full tool-call log and code.

3 Results

3.1 BBN and relic-density allowed region

The server null ($\sin^2 2\theta \to 0$) predicts $Y_p = 0.24682$, D/H$\times 10^5 = 2.4433$, $N_{\rm eff} = 3.0440$, $\chi^2 = 2.77$, $\Delta\chi^2 = 0$, which is $+0.48\sigma$ high in Y_p and -4.15σ low in D/H versus the server baselines, but allowed at 95% ($\Delta\chi^2 < 3.84$). Versus the brief baselines $Y_p = 0.245 \pm 0.003$, D/H$= 2.53 \pm 0.03 \times 10^{-5}$ the offsets are $+0.61\sigma$ and -2.89σ. The offset between brief and server baselines is 0.0001 in Y_p (0.025σ) and 0.047×10^{-5} in D/H (1.88σ on the server error). Linearized slopes about the Standard Model are $dY_p/dN_{\rm eff} = 0.01351$ and $d{\rm D/H}_5/dN_{\rm eff} = 0.3308$, where ${\rm D/H}_5 = {\rm D/H} \times 10^5$. The BBN 95% limit from Y_p/D/H alone is $\Delta N_{\rm eff} < 0.548$ ($\Delta\chi^2$

versus Standard Model > 3.84), quoted also as < 0.55 and < 0.61 depending on baseline and whether $\Delta\chi^2$ is versus Standard Model or best fit. A fully thermalized sterile with $N_{\rm eff} = 4.044$ ($\Delta N_{\rm eff} = 1$) predicts $Y_p = 0.25956$ and D/H$= 2.7769 \times 10^{-5}$, 3.7σ above Y_p, with $\Delta\chi^2 = 23.1$ excluded. Planck $N_{\rm eff} = 2.99 \pm 0.17$ gives $\Delta N_{\rm eff} < 0.33$ at 1σ and < 0.61 at 95% (approximately 2σ), comparable to BBN, while CMB-S4 $\sigma = 0.03$ will give $\Delta N_{\rm eff} < 0.06$.

Figure **??** shows the BBN impact versus $N_{\rm eff}$ with observed Y_p and D/H bands and the 95% limit.

The figure displays Y_p and D/H as functions of $N_{\rm eff}$ with the server baselines overplotted, illustrating the slopes above and that the Standard Model point lies within the Y_p band but low in D/H, with the $\Delta N_{\rm eff} < 0.548$ bound set by the combined χ^2.

DW grids demonstrate that BBN excludes only large mixings. A 7 mass $\times 10$ mixing grid at $1, 3, 7.1, 10, 20, 50, 100$ keV $\times 10^{-15}$ to 10^{-6} gave 12/70 BBN-excluded ($\Delta\chi^2 > 3.84$) and 31/70 overclosing ($\Omega_s h^2 > 0.12$), with e.g. 1 keV/10^{-6} giving $\Delta N_{\rm eff} = 1.0$, $\Omega_s h^2 = 13.84$, $\Delta\chi^2 = 23.1$ excluded, while 7.1 keV/10^{-10} gives $\Delta N_{\rm eff} = 6.2 \times 10^{-4}$, $\Omega_s h^2 = 0.047$ allowed. A 6400-point scan 1–100 keV $\times 10^{-15}$–10^{-6} gave 862/6400 BBN-excluded and 2752/6400 overclosing, with BBN 95% upper limit $\sin^2 2\theta_{\rm max}$ falling from 3.5×10^{-7} at 1 keV to 8.9×10^{-9} at 100 keV (intermediate 1.23×10^{-7} at 3.6 keV, 7.26×10^{-8} at 6.85 keV, 4.29×10^{-8} at 13.8 keV, 2.54×10^{-8} at 26.2 keV, 1.5×10^{-8} at 49.7 keV). A 169-point grid 1 eV–100 keV $\times 10^{-13}$–0.1 gave 70 BBN-excluded and 93 overclosing, with a 10000-point scan giving 3907/10000 BBN-excluded and 5125/10000 overclosing, and $\sin^2 2\theta_{\rm max}$ from 1.15×10^{-3} at 1 eV to 9.3×10^{-9} at 100 keV (1.5×10^{-7} at 3.4 keV, 3.8×10^{-8} at 17.5 keV). A 400-point 1–100 keV $\times 10^{-15}$–10^{-6} scan gave 59/400 BBN-excluded and 172/400 overclosing, with $\sin^2 2\theta_{\rm max}$ from 3.36×10^{-7} at 1 keV to 4.28×10^{-9} at 100 keV. A 143-point 1 eV–100 keV $\times 10^{-13}$–0.1 grid gave 65 BBN-excluded and 89 overclosing. At keV masses relevant to X-ray searches, BBN allows all points with $\sin^2 2\theta \lesssim 10^{-10}$: e.g. 7.1 keV/$10^{-10}$ $\Delta N_{\rm eff} = 0.0006185$, $\Omega_s h^2 = 0.04715$, $\Delta\chi^2 = -0.01$ allowed; 10 keV/10^{-10} $\Delta N_{\rm eff} = 8.1 \times 10^{-4}$, $\Omega_s h^2 = 0.087$ allowed; 20 keV/10^{-10} $\Delta N_{\rm eff} = 0.001416$, $\Omega_s h^2 = 0.3041$ overcloses but BBN allowed; 50 keV/10^{-10} $\Delta N_{\rm eff} = 0.00295$, $\Omega_s h^2 = 1.58$ overcloses but BBN allowed. A 21-point 5–70 keV $\times 10^{-12}$–10^{-10} grid gave 0/21 BBN-excluded and 7/21 overclosing, with $\Delta N_{\rm eff} = 4.7 \times 10^{-6}$ to 0.0039. Varying $\omega_b h^2$ from 0.019 to 0.0257 at fixed $N_{\rm eff} = 3.044$ shifts Y_p 0.24525 $\to$ 0.24824 and D/H 3.1843 $\to 1.9422 \times 10^{-5}$ with absolute χ^2 96.12 $\to$ 87.27 versus 2.77 at fiducial 0.02237, while τ_n 866.4 $\to$ 890.4 s shifts Y_p 0.24433 $\to$ 0.24930, demonstrating baseline dependence of absolute χ^2 but robustness of $\Delta\chi^2 > 3.84$.

The BEST short-baseline best fit $\Delta m^2 = 1.3$ eV^2, $\sin^2 2\theta = 0.42$, $m_s \approx 1.14$ eV gives $\Delta N_{\rm eff} = 1.0$, $N_{\rm eff} = 4.044$, $Y_p = 0.25956$, D/H$= 2.7769 \times 10^{-5}$, $\Delta\chi^2_{\rm BBN} = 23.10$ excluded at 95%, $\Omega_s h^2 = 29.3$ overclosing, in 6.2σ tension with Planck $N_{\rm eff} = 2.99 \pm 0.17$ and 33σ with CMB-S4 $\sigma = 0.03$. The server eV partial thermalization is crude, giving $\Delta N_{\rm eff} = 0.40$ at 1 eV^2, 10^{-3} versus full quantum kinetic 0.93.

Figure **??** shows the exclusion plane from a DW scan.

The figure plots BBN-excluded points ($\Delta\chi^2 > 3.84$) and overclosing points ($\Omega_s h^2 > 0.12$) in the $(m_s, \sin^2 2\theta)$ plane, with the BBN boundary lying one to two orders of magnitude above the $\Omega_s h^2 = 0.12$ contour at keV masses, confirming that overclosure is the binding BBN-related bound for DW.

3.2 X-ray radiative decay constraints

The decay-rate formula was verified to 1–2% against the server: at 7.1 keV/7×10^{-11} predicted 1.74×10^{-28} s^{-1} versus tool 1.72×10^{-28} s^{-1} (1.3%), at 7.1 keV/10^{-10} 2.49×10^{-28} versus 2.46×10^{-28} (1.2%), at 10 keV/10^{-10} 1.38×10^{-27} versus 1.36×10^{-27} (1.4%). Benchmarks for five masses at three mixings give $E_\gamma = m_s/2 = 3.55, 5, 10, 20, 25$ keV, $\Gamma = 2.46 \times 10^{-28}$

(7.1 keV), 1.36×10^{-27} (10 keV), 4.36×10^{-26} (20 keV), 1.39×10^{-24} (40 keV), 4.25×10^{-24} s^{-1} (50 keV) at $\sin^2 2\theta = 10^{-10}$, scaling as $(m_s/\mathrm{keV})^5$, with lifetimes 4.07×10^{27} to 2.35×10^{23} s at 10^{-10}, 4.07×10^{29} to 2.35×10^{25} s at 10^{-12}, and 4.07×10^{31} to 2.35×10^{27} s at 10^{-14}. DW relic densities at 10^{-10} are $\Omega_s h^2 = 0.047$ (7.1 keV), 0.087 (10 keV), 0.304 (20 keV overcloses), 1.06 (40 keV overcloses), 1.58 (50 keV overcloses), with $\Delta N_{\mathrm{eff}} = 0.0006$–$0.0029$ and $\Delta\chi^2 < 0$ (no BBN exclusion) for all fifteen points.

Built-in X-ray checks give: 7.1 keV/7×10^{-11} ($E = 3.55$ keV, $\tau = 5.82 \times 10^{27}$ s, whole-halo fluxes 3.28 ph s^{-1}cm^{-2} Milky Way center, 4.8×10^{-4} M31, 2.5×10^{-5} Perseus) excluded by NuSTAR M31 Roach et al. 2020 and XMM blank-sky Dessert et al. 2020; 10 keV/10^{-10} ($E = 5$ keV, $\tau = 7.35 \times 10^{26}$ s, Perseus 1.4×10^{-4}) excluded; 20 keV/10^{-10} ($E = 10$ keV, $\tau = 2.30 \times 10^{25}$ s, Perseus 2.24×10^{-3}) excluded; 40 keV/10^{-10} ($E = 20$ keV, $\tau = 7.18 \times 10^{23}$ s, Perseus 3.58×10^{-2}) and 50 keV/10^{-10} ($E = 25$ keV, $\tau = 2.35 \times 10^{23}$ s, Perseus 8.74×10^{-2}) not excluded by tabulated limits within 30% mass. At $\sin^2 2\theta = 10^{-12}$ all masses are BBN-allowed and not overclosing ($\Omega_s h^2 = 0.00047$–0.0158) with lifetimes 4.07×10^{29} to 2.35×10^{25} s, requiring external limits. Contrasting high-mass points 40 keV/3×10^{-14} ($E = 20$ keV, $\Gamma = 4.18 \times 10^{-28}$ s^{-1}, $\tau = 2.39 \times 10^{27}$ s, 7.58×10^{19} yr) and 10 keV/10^{-12} ($E = 5$ keV, $\Gamma = 1.36 \times 10^{-29}$ s^{-1}, $\tau = 7.35 \times 10^{28}$ s) are not excluded by built-in limits, confirming BBN negligible at low mixing where X-ray/SPI dominates. At small mixing the X-ray ceiling is mass-dependent: 7.1 keV/10^{-12} ($E = 3.55$ keV, $\tau = 4.07 \times 10^{29}$ s, M31 6.9×10^{-6}) not excluded, while 20 keV/10^{-12} ($E = 10$ keV, $\tau = 2.30 \times 10^{27}$ s, M31 4.3×10^{-4}) remains excluded by NuSTAR M31 even at 10^{-12}, and 40 keV/10^{-14} ($E = 20$ keV, $\tau = 7.18 \times 10^{27}$ s) not excluded. The DW $\Omega_s h^2 = 0.12$ line at $6, 7.1, 10, 20$ keV requires $\sin^2 2\theta \sim 3.4 \times 10^{-10}, 2.5 \times 10^{-10}, 1.4 \times 10^{-10}, 3.9 \times 10^{-11}$ (server) and is excluded by built-in limits at 6–20 keV, while at 40 keV (1.1×10^{-11}, $E = 20$ keV, $\tau = 6.35 \times 10^{24}$ s) it is not excluded by tabulated limits but overcloses at 10^{-10}.

External literature limits were only partially quantified. A contested finding from arXiv:2510.24560 states XRISM Resolve stacked 3.75 Ms of ten galaxy clusters over 2.5–15 keV rest frame found no unidentified lines, with 3σ upper limit $\Gamma \sim 1.0 \times 10^{-27}$ s^{-1} at $m_s \sim 7.1$ keV ($E = 3.55$ keV), 3–4 times stronger than Hitomi Perseus but 5 times above the Bulbul et al. 2014 XMM detection, converting to $\sin^2 2\theta \lesssim 4 \times 10^{-10}$ (XRISM alone) via the server rate 2.46×10^{-28} s^{-1} at 10^{-10}. A contested finding from arXiv:2405.17861 states Krivonos et al. use 11 years of NuSTAR stray-light data on the Milky Way to set 95% CL upper limits on $\sin^2 2\theta$ for $m_s \sim 6$–40 keV (photon 3–20 keV), plotted in their Fig. 3 over 10^{-15} to 10^{-10}, reaching the bottom edge of realistic production models, with $\Gamma = 1.36 \times 10^{-22} (m_s/\mathrm{keV})^5 \sin^2 2\theta$ s^{-1} (1.4% below 1.38×10^{-32} normalized form). The server rate agrees to $\sim 1.5\%$.

Figure **??** shows the NuSTAR stray-light limits.

The figure presents 95% upper limits on $\sin^2 2\theta$ versus m_s from the NuSTAR unfocused stray-light spectrum at $|b| > 3^\circ$ (117 Ms per FPM) together with previous NuSTAR 2020, 2023 and XMM 2021 constraints and theoretical lower bounds, with a borderline excess at 40 keV noted as an instrumental absorbed stray-light effect. Extracted from arXiv:2405.17861.

Figure **??** shows X-ray line detections versus constraints.

The figure displays $1, 2, 3\sigma$ regions for the 3.55 keV line from MOS (blue) and PN (red) stacked clusters, Perseus core-removed (green), M31 (orange) and Galactic Center, versus 90% constraints from Chandra M31, stacked dwarfs and Suzaku Perseus. Extracted from arXiv:1705.01837.

3.3 Production mechanisms and structure formation

Four arXiv identifiers were verified as real papers: arXiv:2507.18752 (Vogel, Garcia Escudero, Froustey, Abazajian 2025, Shi–Fuller revival), arXiv:2502.17374 (Gorbunov, Kalashnikov, Krugan 2025, flavor-dependent asymmetry), arXiv:2601.03346 (Cline and Xu 2026, reheating/inflaton-

decay with branching ratio $\lesssim 10^{-4}$) and arXiv:2609.31340 (Gorbunov and Kalashnikov 2026, saturation at high flavor asymmetry). A contested finding from the full text of arXiv:2507.18752 states $L \gtrsim 0.5$ at high temperature opens viable space at $m_s \gtrsim 10$ keV and $\sin^2 2\theta \lesssim 10^{-14}$, compatible with NuSTAR $\sin^2 2\theta \lesssim 10^{-12}$ at 6–40 keV and INTEGRAL/SPI $\sin^2 2\theta \lesssim 3 \times 10^{-14}$ for $m_s \gtrsim 40$ keV (lifetimes $\lesssim 10^{20}$ s excluded), with structure bounds $m_{\rm th} > 8.4$ keV (galaxy counts, 95% CL), > 8.5 keV (counts+strong lensing) and Lyman-alpha preference $m_{\rm th} = 4.1^{+33.3}_{-1.5}$ keV. The server cannot test this claim: its SF mode caps L at 0.1, its SF $\Delta N_{\rm eff}$ is heuristic $0.1\times$ DW, and asymmetry is not fed into the BBN network. At $10, 20, 30$ keV with $\sin^2 2\theta = 10^{-14}$ it gives $\Delta N_{\rm eff} \sim 10^{-8}$ and $\Delta\chi^2 = 0$ allowed, with $Y_p = 0.24682$, D/H$= 2.4433 \times 10^{-5}$ at 20 keV, while SF $\Omega_s h^2$ values are $0.122, 0.490, 1.10$ for $L = 10^{-3}$ and $1.22, 4.90, 11.0$ for $L = 0.1$, all flagged overclosing, demonstrating the heuristic does not capture resonant enhancement.

DW versus SF BBN contrast is stark: same $(m_s, \sin^2 2\theta)$ points BBN-excluded under DW become allowed under SF heuristic. For $m_s = 7100$ eV, $\sin^2 2\theta = 10^{-7}$, DW gives $\Delta N_{\rm eff} = 0.6185$, $\Delta\chi^2 = 5.89$ excluded versus SF 0.06185, $\Delta\chi^2 = -0.66$ not excluded; for 100 eV, 10^{-5}, DW 1.0 excluded versus 0.1 not excluded. In a 24-point eV–keV grid 14/24 DW points are BBN-excluded ($\Delta N_{\rm eff}$ up to 1.0, $\Delta\chi^2 = 23.1$) versus 0/24 SF points at $L = 0.01$ and 0.1. At 20 keV/10^{-14} DW $\Delta N_{\rm eff} = 1.42 \times 10^{-7}$, $\Omega_s h^2 = 3.04 \times 10^{-5}$ allowed versus SF 1.42×10^{-8}, $\Omega_s h^2 = 4.90$ overcloses by $40\times$; at 50 keV/10^{-14} DW 1.58×10^{-4} versus SF 30.6 overcloses by $255\times$. At $10, 20, 30$ keV/10^{-14} SF gives $\Delta N_{\rm eff} = 8.1 \times 10^{-9}, 1.42 \times 10^{-8}, 1.96 \times 10^{-8}$ allowed but $\Omega_s h^2 = 1.22, 4.90, 11.0$ overclosing, while DW gives 8.1×10^{-8}, 8.7×10^{-6} allowed. The server SF L cap 0.1 therefore cannot test the $L > 0.5$ viable region where recent BBN analyses allow $L > 1$ at $T > 20$ MeV and washout before BBN reopens parameter space.

For DW at small mixing $\sin^2 2\theta = 10^{-12}$ to 10^{-14}, $\Omega_s h^2 \ll 0.12$ (e.g. 3.5×10^{-4} at 6 keV/10^{-12} down to 10^{-4} at 40 keV/10^{-12}, 3.5×10^{-6} at 6 keV/10^{-14}), so DW cannot make dark matter at X-ray allowed mixing; SF resonant with $L \sim 0.1$–1 or inflaton-decay with branching ratio $\lesssim 10^{-4}$ is required. Web snippets confirm inflaton-decay produces cold sterile dark matter during reheating at vanishing mixing, probing $T_{\rm RH}$, and that NuSTAR Galactic bulge 190 ks at $\sim 10^\circ$ off-plane provides complementary limits to M31 1.2 Ms.

Structure-formation bounds from 2024–2026 snippets give thermal-relic $m_{\rm th} > 6.5$ keV (Milky Way satellites DES+PS1, Nadler et al. 2024), > 6.1 keV (strong lensing, 2024) and Lyman-alpha $m_{\rm th} \sim 5$–5.7 keV. Via the Viel et al. DW mapping $m_s = 4.43$ keV$(m_{\rm th}/\text{keV})^{4/3}$ this requires DW $m_s > 53.7$ keV (satellites), > 49 keV (lensing), > 29–41 keV (Lyman-alpha 4.1–5.3 keV), consistent with earlier > 28–48 keV (Lyman-alpha) and > 76 keV (satellites 8.4–8.5 keV, giving > 76 keV and > 78 keV). SF resonant and inflaton-decay production are colder (lower $\langle p\rangle/T$ by ~ 1.5–$3\times$), so same m_s corresponds to larger effective $m_{\rm th}$ and bounds shift down, reopening $m_s \sim 10$–20 keV at $\sin^2 2\theta < 10^{-12}$ where X-ray limits also weaken.

Figure **??** shows the warm dark matter transfer function.

The figure displays the WDM transfer function $T(k)$ for pure ΛWDM with $T_x/T_\nu = 0.5, 0.3, 0.25, 0.226, 0.2$ corresponding to $m_x = 92, 427, 738, 1000, 1441$ eV and for mixed ΛCWDM models, with numerical curves and analytical fits. Extracted from arXiv:astro-ph/0501562.

Figure **??** shows the flux power spectrum suppression.

The figure shows fractional difference of flux power spectrum for ΛWDM versus ΛCDM at $z = 4, 3, 2.5$ for thermal masses $0.5, 1, 2, 4, 8$ keV, illustrating increasing suppression at small scales for lighter masses. Extracted from arXiv:astro-ph/0501562.

Figure **??** shows the bias function.

The figure plots bias $b^2(k)$ between flux power spectrum and linear matter power spectrum for ΛCDM and ΛWDM with $m_{\rm WDM} = 1$ keV, with shaded analysis range. Extracted from arXiv:astro-ph/0501562.

3.4 Normalization and robustness

The server DW relic density scales as $\Omega_s h^2 = 0.1\,(\sin^2 2\theta/10^{-9})(m_s/3\text{ keV})^{1.8}$, verified by 3 keV/$10^{-9}$ $\to$ 0.100, 10 keV/10^{-9} $\to$ 0.873 (ratio 8.73 = $(10/3)^{1.8}$), but normalization is 3$\times$ higher than the brief $0.1\,(\sin^2 2\theta/3\times10^{-9})(m_s/3\text{ keV})^{1.8}$. The published Abazajian 2017 review arXiv:1705.01837 Eq. 37 fit $m_s \sim 3.40\text{ keV}(\sin^2 2\theta/10^{-8})^{-0.615}(\Omega_{\rm DM}/0.26)^{0.5} \times 0.527\,\mathrm{erfc}[-1.15\,(T_{\rm QCD}/170\text{ MeV})^{2.15}]$ inverts to $\sin^2 2\theta \sim 1.23\times10^{-8}$ (3 keV), 3.0×10^{-9} (7.1 keV), 1.73×10^{-9} (10 keV), 5.6×10^{-10} (20 keV), 1.26×10^{-10} (50 keV) for $\Omega_s h^2 = 0.12$, while the server gives $1.2\times10^{-9}, 2.55\times10^{-10}, 1.37\times10^{-10}, 3.9\times10^{-11}, 7.6\times10^{-12}$, a factor 10$\times$ (3 keV) to 17$\times$ (50 keV) lower mixing. Equivalently $\Omega_s h^2 \approx 1.4\times10^7(m_s/1\text{ keV})^{1.8}\sin^2 2\theta$ in the server versus $1.74\times10^8(m_s/1\text{ keV})^{1.8}\sin^2 2\theta$ in Abazajian, factor $\sim$ 12.4. At 7.1 keV/7×10^{-11} server $\Omega_s h^2 = 0.033$ versus 0.42 expected, shifting the $\Omega_s h^2 = 0.12$ contour to $\sim 12\times$ larger mixing. A contested finding notes that at 7.1 keV the DW verdict versus XRISM $\sin^2 2\theta \lesssim 4\times10^{-10}$ depends on normalization: server line 2.55×10^{-10} survives XRISM, while brief 7.65×10^{-10} and published 3.0×10^{-9} are excluded by $\sim 7.5\times$.

BBN robustness checks using linearized slopes about server Standard Model values give 95% upper bounds $\Delta N_{\rm eff} < 0.32$ (Y_p-only), < 0.58 [0.44 versus best fit] (D/H-only), < 0.47 [0.37] combined for brief data, and < 0.46, < 0.66 [0.46], < 0.61 [0.43] for server data, with server tool reporting 0.548. D/H alone prefers $\Delta N_{\rm eff} = 0.26$ (brief) and the server notes $dY_p/d\xi_e \sim -0.25$ for joint $(\Delta N_{\rm eff}, \xi_e)$ bounds, but no emulator call probes $L > 0.5$ where $\xi_e \sim 0.01$–0.06 could compensate $\Delta N_{\rm eff} \sim 0.5$–$1.0$.

4 Discussion

The combined BBN and X-ray analysis establishes a clear hierarchy: BBN alone is not binding for keV DW dark matter at X-ray relevant mixings, with $\Delta N_{\rm eff} \ll 0.548$ and $|\Delta\chi^2_{\rm BBN}| < 0.15$ along the $\Omega_s h^2 = 0.12$ contour, while overclosure and X-ray non-detections are binding. The DW $\Omega_s h^2 = 0.12$ line lies at $\sin^2 2\theta \sim 10^{-10}$–$10^{-11}$ (server) or $\sim 10^{-9}$–10^{-10} (published), both above NuSTAR $\lesssim 10^{-12}$ at 6–40 keV and SPI $\lesssim 3\times10^{-14}$ above 40 keV, and above XRISM $\Gamma \sim 10^{-27}$ s^{-1} ($\sin^2 2\theta \lesssim 4\times10^{-11}$–$4\times10^{-10}$ at 7.1 keV depending on conversion). Structure formation then requires $m_s > 28$–76 keV for DW, where X-ray limits are strongest, leaving no surviving DW all-dark-matter interval. This agrees with the literature consensus that minimal DW is essentially excluded and that non-standard production is required.

Tensions and limitations are dominated by incomplete external validation. Several findings remain contested because verbatim quotes or figure readings were not obtained: the $L \gtrsim 0.5$ opening of $m_s \gtrsim 10$ keV, $\sin^2 2\theta \lesssim 10^{-14}$ (arXiv:2507.18752), the XRISM 3σ $\Gamma \sim 10^{-27}$ s^{-1} limit (arXiv:2510.24560), and the NuSTAR 6–40 keV $\sin^2 2\theta \lesssim 10^{-12}$ and SPI $\lesssim 3\times10^{-14}$ ceilings. The server SF heuristic (0.1$\times$ DW, $L \le 0.1$, no ξ_e) is unphysical at $L = 0.1$ (overclosing by 40–255$\times$ at 10^{-14}) and cannot test the $L > 0.5$–1 regime where recent BBN analyses allow $L > 1$ at $T > 20$ MeV to wash out before BBN. The DW normalization offset factor ~ 12 means server-based DW statements are not conservative; using the published normalization, DW at 7.1 keV is excluded by XRISM alone.

Dead ends observed include: searches for 2024–2026 eROSITA and INTEGRAL/SPI line limits returned only older or unrelated papers (1505.07829, 1702.08430, muon capture, 2008 heating); targeted INTEGRAL/SPI searches at $m_s \ge 40$ keV returned no primary 2024+ source, leaving the SPI entry reliant on the arXiv:2507.18752 quote; web searches for XRISM and NuSTAR suffered DNS resolution failures for `wt.wikipedia.org.cels.anl.gov`; tool budget exhaustion prevented `read_arxiv_chunk`, `cite`, `get_paper_figures` and additional `predict_sterile_nu_xray_line` calls, so no verbatim Abazajian Eq. 37 or Viel coefficient quotes, no numerically read NuSTAR Fig. 3 observed versus expected limits at $7, 10, 20, 40$ keV,

no XRISM Fig. 3, no INTEGRAL/SPI primary (Boyarsky diffuse, Roach SPI) to resolve the 50 keV lifetime mismatch (tool $\tau = 7.84 \times 10^{26}$ s at 3×10^{-14} versus quoted 10^{20} s), no corrected five-row X-ray limit table with provenance per mass, no per-mechanism viability table with Viel translation at 7.1/10/20/50 keV, no exclusion-plane plot distinguishing tool-internal versus literature limits with $\Omega = 0.12$ overlay, and no robustness reruns varying $\omega_b h^2$ and τ_n. The critic verdicts reflected this: initial plans were marked revise, subsequent waves continue with contested findings, and final reports were marked revise until the last wave was marked ok.

5 Conclusions

Established with high confidence from server calculations is that BBN allows keV sterile-neutrino dark matter at $\sin^2 2\theta \leq 10^{-10}$ ($\Delta N_{\rm eff} = 6 \times 10^{-5}$ to 2.9×10^{-3}, $\Delta\chi^2 < 0$), that DW overclosure is 40× to 5000× stronger than BBN at 1–100 keV, that the radiative decay rate follows $\Gamma = 1.38 \times 10^{-32}\ {\rm s}^{-1} (\sin^2 2\theta / 10^{-10}) (m_s/{\rm keV})^5$ to 1–2%, and that built-in X-ray limits exclude 7.1 keV/7×10^{-11}, 10 keV/10^{-10} and 20 keV/10^{-10} while 40–50 keV/10^{-10} are not excluded by tabulated limits but overclose. The eV BEST point is BBN-excluded ($\Delta\chi^2 = 23.1$) and overcloses. Viable dark matter at X-ray allowed mixings $\sin^2 2\theta \lesssim 10^{-12}$–$10^{-14}$ requires non-minimal production: SF resonant with $L \gtrsim 0.5$–1 at $T > 20$ MeV (arXiv:2507.18752, arXiv:2512.00793), flavor-dependent asymmetries (arXiv:2502.17374), reheating/inflaton-decay with branching ratio $\lesssim 10^{-4}$ (arXiv:2601.03346, arXiv:2601.02069), nuSMEFT freeze-in (arXiv:2409.08968) or time-dependent mass (arXiv:2502.13123), which are colder and relax structure-formation bounds.

What remained open is the quantitative external X-ray limit table with primary figure numbers and verbatim quotes, the factor ~ 12 DW normalization provenance with Abazajian Eq. 37 text, the Viel $m_s = 4.43\ {\rm keV} (m_{\rm th}/{\rm keV})^{4/3}$ coefficient verbatim, and the joint BBN+X-ray exclusion overlay distinguishing BBN versus built-in versus literature limits.

Ranked open directions as listed by the board are:

1. Deepen

Data and code availability

All outputs of this session are collected in the project folder `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray`. The session produced 45 data files; the supplement lists them. Full documentation of how these results were obtained is in the supplementary material (`supplement.pdf`: implementation details, the tool-call log, the research board with every finding and its evidence, figures not shown here, and the code index) and the verbatim transcript (`transcript.pdf`), generated alongside this report.